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Brief Report

Magnetic Resonance Imaging Versus Computed Tomography Guidance for Stereotactic Body Radiotherapy in Prostate Cancer: 2-year Outcomes from the MIRAGE Randomized Clinical Trial

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Abstract

It has been shown that magnetic resonance imaging (MRI) guidance versus computed tomography (CT) guidance for aggressive margin-reduction (AMR) for stereotactic body radiotherapy (SBRT) in prostate cancer reduces acute toxicity, but the longer-term benefits are unknown. We performed a secondary analysis of MIRAGE, a phase 3 randomized clinical trial of MRI-guided SBRT for prostate cancer, to determine whether AMR with MRI guidance significantly reduced 2-yr physician-scored or patient-reported toxic effects in comparison to CT guidance. The cumulative incidence of 2-yr physician-scored toxicity, defined as grade ≥2 genitourinary (GU) and gastrointestinal (GI) toxic effects according to Common Terminology Criteria for Adverse Events v4.03, were lower with MRI guidance. Cumulative incidence rates of late grade ≥2 toxicity at 2 yr with MRI-guided versus CT-guided SBRT were 27% (95% confidence interval [CI] 19–39%)] versus 51% (95% CI 41–63%) for GU toxicity (p = 0.004), and 1.4% (95% CI 0.2–9.6) versus 9.5% (95% CI 4.6–19) for GI toxicity (p = 0.025). Cumulative logistic regression revealed that MRI-guided SBRT was associated with significantly lower odds of a clinically relevant deterioration in bowel function according to the Expanded Prostate Cancer Index Composite-26 score (odds ratio 0.444, 95% CI 0.209–0.942; p = 0.035) and in the Sexual Health Inventory in Men score (odds ratio 0.366, 95% CI 0.148–0.906; p = 0.03). There were no significant differences in the odds of a deterioration for other quality-of-life metrics. These findings support the hypothesis that aggressive planning for margin reduction for prostate SBRT using MRI leads to continued reductions in toxic effects over 2-yr follow-up.

This trial is registered on ClinicalTrials.gov Identifier as NCT04384770.

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ADVANCING PRACTICE

What does this study add?

These secondary analyses of a phase 3 randomized trial support the hypothesis that aggressive margin-reduction (AMR) using magnetic resonance imaging (MRI) to guide stereotactic body radiotherapy (SBRT) for prostate cancer reduces toxicity over the 2 yr following treatment. Cumulative incidence rates for grade ≥2 genitourinary and gastrointestinal toxicity were significantly lower with MRI-guided than with computed tomography–guided SBRT. Clinically meaningful deterioration in patient-reported bowel function and sexual health score was also lower in the MRI-guided SBRT arm. These data suggest that AMR achieved with MRI guidance can reduce post-SBRT toxicity in men with prostate cancer.

Clinical Relevance

The ultimate goal of research in radiation oncology is to maximize treatment delivery to the target while minimizing effects on surrounding structures. The MIRAGE trial is one of few randomized studies on prostate cancer radiation therapy that assesses a new technology in this setting. The investigators randomized patients with clinically localized prostate cancer to stereotactic body radiation therapy with MRI guidance and 2-mm planning target volume margins or the more conventional CT-guidance with 4-mm margins. In this prespecified secondary analysis of the trial two years after treatment the Authors showed that MRI-guided stereotactic body radiation therapy reduces gastrointestinal and genitourinary toxicity, and is associated with less deterioration of bowel and sexual function. This work supports the use of MRI-guided stereotactic body radiation therapy with tight margins for men with clinically localized prostate cancer to reduce post-treatment adverse effects. Associate Editor: Gianluca Giannarini MD.

Patient Summary

To protect surrounding tissue, imaging scans are used to plan small margins around tumors for radiation therapy in prostate cancer. We compared two imaging methods and found that MRI (magnetic resonance imaging) guidance for even smaller margins resulted in lower rates of side effects than the standard CT (computed tomography) guidance. Updated results at 2 years after treatment show that smaller margins with MRI guidance were linked to less deterioration in patient-reported bowel function and sexual health.

Stereotactic body radiotherapy (SBRT) is a standard-of-care treatment for localized prostate cancer [1,2]. The phase 3 MIRAGE trial demonstrated that use of a magnetic resonance imaging (MRI)-guided linear accelerator platform to aggressively reduce planning margins from 4 mm (typical margin with computed tomography [CT]-guided linear accelerator platforms) to 2 mm resulted in significantly lower rates of acute physician-scored grade ≥2 genitourinary (GU) and gastrointestinal (GI) toxicity [3]. The incidence of patient-reported deterioration in urinary and bowel quality-of-life (QOL) measured in the first 3 mo after SBRT was also lower among patients receiving MRI-guided prostate SBRT. However, it is unknown whether these differences persist over longer-term follow-up.

In this prespecified secondary analysis of the single-center phase 3 randomized controlled MIRAGE trial (NCT04384770) [3] we evaluated both physician-scored toxicity and patient-reported QOL measured at the 2-yr time point. We chose 2 yr as this is the time point for primary endpoint analysis in multiple contemporary trials, predicated on the observation that the relative risks of GU and GI toxicity at intermediate follow-up (2 yr) do not significantly change over longer-term follow-up [4].

Patients with localized prostate cancer were randomized 1:1 to receive CT-guided SBRT with a 4-mm planning target volume (PTV) margins or MRI-guided SBRT with 2-mm PTV margins using permuted block randomization (Supplementary material). Participants were stratified by baseline International Prostate Symptom Score (IPSS; ≤15 or >15) [5] and prostate gland volume (≤50 ml or >50 ml) measured via diagnostic MRI. A total of 178 patients were randomized between May 2020 and October 2021 (88 CT and 90 MRI), with 22 patients dropping out before treatment. Thus, 156 patients (77 CT-guided and 79 MRI-guided) received SBRT as randomized. The Consolidated Standards of Reporting Trials (CONSORT) guideline was followed (Supplementary Fig. 1). Dropout rates were equivalent between the arms and reflected patient decision to opt for alternative definitive local therapies after their initial consultation at which trial enrollment was offered.

The primary outcome was acute GU grade ≥2 toxicity (within 90 d of SBRT) according to Common Terminology Criteria for Adverse Events (CTCAE) v4.03. Secondary outcomes included late GU and GI grade ≥2 toxicity and changes in IPSS, Expanded Prostate Cancer Index Composite-26 (EPIC-26) [6], and Sexual Health Inventory for Men (SHIM) [7] scores over time. Clinically relevant deterioration was defined as a decrease of >18, 14, 12 and 24 points in EPIC urinary incontinence, urinary irritative/obstructive symptoms, and bowel and sexual function domains, respectively, an increase of ≥15 points in IPSS, or a decrease of ≥10 points in SHIM score [8]. Patients were evaluated for toxicity and completed QOL questionnaires at baseline and at 1, 3, 6, 9, 12, 18, and 24 mo after SBRT.

The cumulative incidence of late grade ≥2 GU and GI toxicity was estimated using the Kaplan-Meier method as the time to first occurrence from 6 to 24 mo, with comparisons performed via log-rank tests. Mixed-effects logistic regression was used to determine if there were differences in the odds of a clinically relevant deterioration in QOL measures from 6 to 24 mo between the groups. The model included a fixed effect for treatment group and a random effect for patients. Patients with at least one follow-up visit between 6 and 24 mo were included in the analyses.

Baseline patient and treatment characteristics are shown in Supplementary Table 1. Overall, 19% of patients had favorable intermediate-risk disease, 44% had placement of a rectal hydrogel spacer, 27% received nodal radiation, and 26% received a simultaneous integrated boost to the dominant lesion, with no significant differences between the arms for any of these parameters.

MRI-guided SBRT versus CT-guided SBRT was associated with significantly lower cumulative incidence of late grade≥2 toxicity at 2 yr for both GU toxicity (27%, 95% confidence interval [CI] 19–39% vs 51%, 95% CI 41–63%; p = 0.004) and GI toxicity (1.4%, 95% CI 0.2–9.6% vs 9.5%, 95% CI 4.6–19%; p = 0.025; Fig. 1, Supplementary Fig. 2, and Supplementary Tables 2 and 3).

Compliance with QOL survey completion was >83% across all instruments at 24 mo (Supplementary Figs. 3 and 4). The proportion of patients reporting a clinically relevant deterioration in urinary irritation and bowel function at any point between 6 and 24 mo was lower in the MRI-guided SBRT group (14/73 vs 24/68 for urinary irritation, p = 0.031; 19/72 vs 30/71 for bowel function, p = 0.046; Supplementary Table 4 and Supplementary Figs. 5 and 6). There were no significant differences in the proportions of patients reporting clinically relevant deterioration for other QOL metrics.

Cumulative logistic regression revealed that MRI-guided SBRT was associated with significantly lower odds of a clinically relevant deterioration in bowel function (odds ratio 0.444, 95% CI 0.209–0.942; p = 0.035) or SHIM score (odds ratio 0.366, 95% CI 0.148–0.906; p = 0.03; Table 1 and Supplementary Table 5). There were no significant differences in the odds of experiencing a deterioration in other QOL measures.

This secondary analysis of the MIRAGE trial suggests that the significantly better results for acute physician-scored toxicity and patient-reported QOL achieved with aggressive margin reduction using MRI-guided SBRT persists over longer-term follow-up. We showed that for patients receiving MRI-guided SBRT the cumulative incidence of grade ≥2 GU and GI toxicity was significantly lower over 2 yr than for patients receiving CT-guided SBRT. Logistic regression revealed that the odds of experiencing a clinically relevant deterioration in bowel function or SHIM score were significantly lower in the MRI-guided SBRT arm. Much of the benefit can probably be attributed to the margin reduction, with real-time tracking of the prostate itself—a key aspect that helped in reducing the margins—also being important [3,9]. The enhanced imaging capabilities of the MRI-guided platform were used to achieve this margin reduction; in theory, other devices with similar abilities could achieve the same result.

Notably, while the findings for bowel toxicity were robust, those for urinary toxicity and QOL and sexual function were less consistent. The reasons are likely to be multifactorial. Most late grade ≥2 GU toxicity events were related to prescription of an α-adrenergic antagonist; if this medical intervention were effective, then patient-reported QOL measures would not capture a deterioration, whereas physician-rated grade 2 GU toxicity would be recorded. SHIM focuses on erectile function exclusively, while the EPIC-26 sexual function domain also evaluates other aspects of function. It is plausible that erectile function

Table 1 – Logistic regression results for a cumulative change (6–24 mo) in outcome greater than two times the minimum clinically important difference with magnetic resonance imaging versus computed tomography guidance[a]

Outcome	Odds ratio (95% CI)	p value
EPIC-26 urinary incontinence	0.72 (0.31–1.68)	0.4
EPIC-26 urinary irritative symptoms	0.54 (0.25–1.17)	0.12
EPIC-26 bowel function	0.44 (0.21–0.94)	**0.035**
EPIC-26 sexual function	0.68 (0.27–1.69)	0.4
International Prostate Symptom Score	0.71 (0.23–2.15)	0.5
Sexual Health Inventory for Men	0.37 (0.15–0.91)	**0.03**

EPIC-26 = Expanded Prostate Cancer Index Composite-26; CI = confidence interval.

[a] Odds ratios and p values were obtained using a mixed-effects logistic regression model with fixed effects for treatment and baseline score. A random effect for subject was used with an unstructured covariance structure to model within-patient errors. The Kenward-Roger approximation was used to estimate the denominator degrees of freedom.

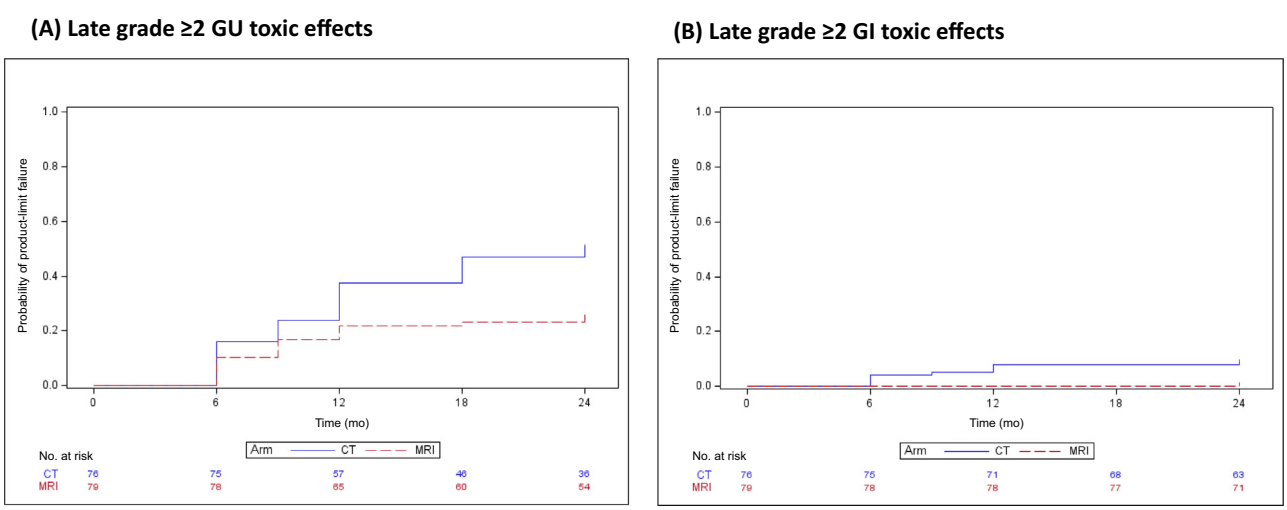

Fig. 1 – Cumulative incidence of late grade ≥2 (A) genitourinary (GU) and (B) gastrointestinal (GI) toxicity, plotted as curves for the probability of product-limit failure for CT-guided and MRI-guided stereotactic body radiotherapy. CT = computed tomography; MRI = magnetic resonance imaging.

may be better spared with tighter planning margins [10], while other adverse effects (such as low ejaculate volume) may be caused by irradiation of the prostate itself and proximal seminal vesicles and thus would not be expected to be impacted by tighter margins. The absolute rate of grade ≥2 GU toxicity in this study, compared to 32% in the PACE-B study, could be attributed to the substantially higher biological dose used in MIRAGE (40 Gy in 5 fractions vs 36.25 Gy in 5 fractions, similar to a 14 Gy dose escalation for late urinary effects).

Finally, the trial was not designed to evaluate the cost-effectiveness of MRI versus CT guidance for SBRT from either a payer or a provider perspective and the inherent value of adopting an MRI guidance system requires careful consideration, particularly given that some QOL metrics did not significantly differ between the arms. However, the results provide proof-of-principle data that aggressive margin reduction does improve multiple measures of urinary, bowel, and sexual QOL.

The primary limitation of this analysis is the lack of blinding, which may have impacted both physician-scored toxicity and patient-reported QOL results. However, as discussed previously, blinding was impractical [3]. Furthermore, the MIRAGE trial was not powered for any of the secondary analyses evaluated here. While the study population was heterogeneous with respect to field size, hydrogel use, and ADT use, we believe that this heterogeneity adds to the generalizability of the results.

In conclusion, our secondary analysis of a randomized trial revealed that aggressive planning for margin reduction with MRI guidance versus CT guidance for prostate SBRT led to lower physician-scored GU and GI toxicity and better bowel and sexual QOL metrics over 2 yr of follow-up.

Author contributions: Amar U. Kishan had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Kishan, Wilhalme, Lamb, Cao, Steinberg.
Acquisition of data: All authors.
Analysis and interpretation of data: All authors.
Drafting of the manuscript: Kishan, Wilhalme.
Critical revision of the manuscript for important intellectual content: All authors.
Statistical analysis: Wilhalme.
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Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eururo.2024.10.026.

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